

Survey of Benchmarking Methods for Quantum Logic Gates

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Abstract

Accurate benchmarking of quantum gates is essential for improving the performance of quantum computing-based technologies. This paper surveys established and emerging methods for characterizing fidelity and error mechanisms. We explore techniques such as Randomized Benchmarking (RB) and Cross-Entropy Benchmarking (XEB). General methods include Quantum Process Tomography (QPT), Gate Set Tomography (GST), and Direct Fidelity Estimation (DFE). Lastly, we provide an overview of two emerging approaches: Leakage Benchmarking and Deterministic Benchmarking.

Introduction

Quantum logic gates are used in quantum computing devices as registers that hold quantum bit values of either $|1\rangle$ or $|0\rangle$, like traditional logic gates, but with an additional complex amplitude associated with each value that accounts for probability.

$$\psi = \alpha |1\rangle + \beta |0\rangle \tag{1}$$

As such, equation (1) above gives us a way to write the state of a qubit. These registers are called qubits and can be imagined as a vector, with quantum gates being transformation matrices that change these qubits while preserving the probability of the vector. Qubits and their probability component are important to quantum computing because they can be acted upon in parallel, like GPU computations, and can use probability to prioritize correct answers. Sorting and searching algorithms have been significantly quicker with the use of quantum logic gates over traditional methods.

The Bloch Sphere is a common way to visualize the state of a qubit (a superposition of a $|1\rangle$ or $|0\rangle$ state), where all pure states are mapped to the surface of a sphere. An observation collapses to either a 1 or a 0 with probability relative to the distance of the vector projected onto the z-axis (used for the $|1\rangle$ and $|0\rangle$ states).

Quantum logic gates acting on a state can be seen as a rotation of a vector from one state to another point on the surface of the sphere, and how accurate this operation is compared to the expected result gives a sense of the performance of the quantum gate.

In benchmarking quantum gates, one must be mindful of both coherent and incoherent sources of error, where coherent error is consistent and comes from errors from the gate operation on the quantum state, and incoherent errors are stochastic and come from inconsistencies with the system itself. In a more technical sense, coherent errors respect the unitary properties of a gate, whereas incoherent errors do not [1]. Coherent errors are seen as over- or under-rotations of a quantum state represented on the Bloch Sphere, and incoherent errors show as the vector of the state in the Bloch Sphere shrinking in magnitude.

Randomized Benchmarking

Randomized benchmarking is important for quantum logic gates because, like traditional logic gates, they can be lossy and imperfect. This randomized methodology of benchmarking follows a Monte Carlo style by measuring the difference between an implemented quantum gate and the ideal quantum gate when injected with random sequences.

The predominant method for benchmarking quantum gates has been the randomized benchmarking (RB) process [1]. This process consists of preparing an initial state, applying a sequence of operations with a quantum gate, and observing the results to see if they match the expected results. The random sequences rely on a group known as the Clifford group, which is shown to be efficient to implement by Gottesman [2] (Gottesman-Knill Theorem).

The Clifford Group encompasses the set of operations that map members of the Pauli group onto itself.

$$C_n = \{C : CPC^\dagger \in P_n, \forall P \in P_n\} \quad (2)$$

Where the Pauli Group is defined as the set of the identity matrix and all the Pauli matrices, and their products with ± 1 and $\pm i$. These gates form the basis of how the RB process is performed. Any gate of this group can be implemented using Hadamard, Phase,

and CNOT gates [2]. The conventional process of RB starts with selecting a depth to uniformly and randomly generate a sequence of Clifford gates, then a final Clifford gate, chosen to invert the result of all preceding gates, is calculated. Aaronson and Gottesman [3] demonstrate how the inversion gate can be efficiently calculated and implemented.

$$S_n = C_1 C_2 \dots C_n \quad (3)$$

$$C_{n+1} = C_1^\dagger C_2^\dagger \dots C_n^\dagger \quad (4)$$

$$S C_{n+1} = I \quad (5)$$

A sequence is generated, and the final gate is calculated to finish the sequence where the expected result of the total sequence is equivalent to the identity matrix. With this inversion gate, the result of this string of operations is equivalent to applying the identity matrix to a prepared starting state. Applying this sequence will theoretically return the initial state of the system to itself. This process is then repeated many times, with new randomly generated sequences and a survival probability is measured for a certain depth [4]. This serves to characterize the performance of the quantum gate as it implements the whole set of Clifford gates. General performance is quantified as the general survival rate of the initial state, and the performance of the quantum gate as it implements the set of Clifford Gates can be empirically determined.

This simple method reduces the performance of the gate to a single number, which does not distinguish between coherent and incoherent errors. However, RB will often underestimate the coherent error of the system, as the coherent errors add in phase, and the random selection of gates can diminish the observed error, whereas incoherent errors add as probabilities, so the results of RB track closer to the incoherent errors of the system [1]. Additionally, the general RB process does not quantify the performance of a specific gate, but rather the expected performance of the quantum gate as it implements the Clifford group.

The simplicity and efficiency of the process lend to its widespread adoption. There are some sources of errors with this method. State Preparation and Measurement errors (SPAM) do not involve the quantum gate itself, but account for inaccuracies within the preparation of the input and output of the gate. Though If the experiment is carefully conducted it can be factored out in the results [1]. An additional source of error is the individual noise factor of each gate, which is the difference in the inherent noise behavior of each gate.

Other schemes of RB have been developed to address some of the issues. Such as inter-

leaved RB (IRB), which aims to isolate the performance of a specific Clifford gate [5]. The process differs that after every gate in the random sequence, a specific Clifford gate is placed in the sequence.

$$S_n = C_1 C C_2 C C_3 \dots C_n \quad (6)$$

$$C_{n+1} = C_1^\dagger C^\dagger C_2^\dagger C^\dagger C_3^\dagger \dots C_n^\dagger \quad (7)$$

Both the random benchmark sequence and this interleaved sequence are then performed. The results of the interleaved sequence are compared to the fully randomized sequence, and the effects of the interleaved gate are determined [5].

Another popular form of RB is Unitary or Purity Randomized benchmarking (URB), where the decay of the state from a pure state to a mixed one is measured [6]. This scheme aims to measure the coherent error by evaluating the unitarity of the output of an RB experiment defined by J. Wallman et al. in [6]. The unitarity of the gate is closely related to the purity of the state measured, and can be seen as a measurement of how mixed the results of the RB experiment have become. J. Wallman et al. [6] describe a protocol similar to the first RB process explained, but it differs in that the result of the random sequence does need not return the state back to the original prepared state, and an expectation value for the sequence of Clifford gates generated is calculated. Then the expectation of over a noisy gate is shown to be an exponential function of the length of the random sequence as well as the unitarity value of the gate. Because of this relationship, multiple trials of the random benchmark sequence with varying lengths can be performed and fit to this exponential to empirically calculate the unitarity of the sample, where a unitarity close to one characterizes the error as more coherent [6].

Other forms of RB have also been developed with their own ramifications, J. Miguel-Ramiro et al. in [7] propose a process called Coherent Randomized Benchmarking, where a group of Clifford gates is of interest, and posit that by building randomized sequences using groups of gates, rather than individual gates, an error term can be much more efficiently measured for the groups of operations compared to using the original RB process. Other processes exploring randomized sequences outside of the Clifford group have also been proposed [8]. Overall, the applicability of the RB process, as well as novel implementations of the process, is a relatively active area of ongoing research.

In conclusion, random benchmarking is an efficient and simple way to measure the performance of a quantum gate, but it does have some downsides; with not explicitly measuring

coherent vs incoherent errors and often underestimating the coherent errors with the gate, as well as not directly measuring the performance of specific gates. Other methods of applying the RB process that have been developed to address these shortcomings, such as the Interleaved Randomized Benchmark, or the Unitary Randomized Benchmark, which help with characterizing a specific gate, or measuring the coherent errors of a gate, respectively. Current research in the field primarily focuses on the scalability of these gates and creating consistent and reliable methods for measuring the performance of quantum logic gates under noisy, real-world conditions. Quantum computers must first have a low error rate before being expanded upon and scaled up, which is what drives the importance of these benchmarking methods.

In a similar vein to random benchmarking, Cross-Entropy Benchmarking aims to calculate the fidelity of a quantum computer by implementing a random quantum circuit many times and collecting samples from each run. The expectation of the samples are then compared to the expectation values of the quantum circuit if an ideal quantum computer was used [9]. By crossing certain thresholds of accuracy demonstrated by S. Boixo et al. in [9] it can be seen that the process to obtain the samples would be more efficient using this quantum computer compared to equivalent classical methods (coined quantum supremacy, a problem that is more efficient to solve on a quantum computer compared to a classical one).

Next, we examine general benchmarking methods that are applicable not only to obtain an aggregate metric but also to model the system overall. At the expense of a more resource-intensive computation, we get a more detailed story about gates or gate sets with methods like Quantum Process Tomography (QPT), Gate Set Tomography (GST), or Direct Fidelity Estimation (DFE).

Quantum Process Tomography

Quantum Process Tomography aims to reconstruct the entire quantum process of a gate or operation. In QPT, one prepares a set of different input states that form a tomographically complete basis, applies the noisy quantum gate to each, and measures the outputs in various bases [10]. By collecting all input-output correlations in different scenarios, QPT yields an estimate of the process matrix describing the gate’s effective quantum operations. In other words, QPT provides the complete description of “what the gate does,” without looking at its intended purposes, allowing calculation of any fidelity or error metric [10]. We can also infer that it is the most detailed characterization because it produces a map that predicts the outcome of any input state.

Such detailed measurements obviously require strong assumptions. QPT assumes that error-free State Preparation and Measurement (SPAM) is achieved [11]. In the current state of research, it is hard for researchers to eliminate SPAM errors, and research methods are still being proposed on the characterization of these errors in a general quantum computing system [12]. In practice, if the preparation or measurement processes are themselves error-prone, QPT will attribute those errors to the gate, leading to incorrect process estimates [10]. QPT also assumes the noise is stationary and Markovian during data collection, so that each mapping is well-defined and does not fluctuate with time. Violating these assumptions can cause QPT to yield unphysical or inconsistent results, nullifying the measurements. Although these assumptions are being challenged with new methods by White et al., at the current stage, these QPT assumptions render the process impractical [13].

The resource cost of QPT grows exponentially with size. To characterize a d -dimensional quantum operation (for n qubits, $d = 2^n$), one needs on the order of d^2 independent input states and d^2 measurement settings to fully probe the process, leading to d^4 outcome probabilities to estimate [14]. Even with possible optimizations, the number of experiments and the classical post-processing scale poorly as n increases [15, 16]. In practice, QPT has been demonstrated only for small systems (up to 2–3 qubits) because of this exponential overhead [16]. Thus, while QPT is comprehensive, it is not scalable for large-scale systems or circuits.

The benefit of QPT is its ability to diagnose all types of errors. In principle, QPT is sensitive to all types of errors in the sense that a full process reconstruction can reveal any deviation of the channel from the ideal unitary, including stochastic errors, unitary errors, leakage to outside states, etc. Because it returns the complete process matrix, one could diagnose specific error channels from the reconstructed map. However, due to SPAM sensitivity, some error types can be misidentified or invisible. For example, if there is consistent measurement bias or state-preparation error, QPT will “see” an error but incorrectly assign it to the gate’s process instead. There are some methods to mitigate SPAM errors, such as a multipass QPT system, but the resource requirements of such systems skyrocket [17].

If done carefully on a single or two-qubit gate, the QPT data can guide calibration improvements by revealing the specific error mechanism. However, because of the trust assumptions required, QPT by itself is seldom used for routine large-scale tuning. Its most actionable use today is as a validation tool for small systems or as a component in more advanced self-calibrating methods such as GST. In summary, QPT provides the most granular insight into a gate’s behavior, which in principle can guide targeted fixes, but its practicality is confined to small, well-controlled scenarios.

Gate Set Tomography

Gate Set Tomography produces a complete characterization of an entire gate set, including the quantum processes for each logic gate and the state preparation and measurement operations [10]. Rather than estimating one gate in isolation, GST simultaneously fits all operations to find a physical model that best explains the experimental outcomes. The end result is a calibrated “gate set”, which is a collection of CPTP maps for the initially prepared state, the POVM measurement, and each gate in the set [18]. Essentially, GST extends process tomography to yield all the systematic error parameters of the quantum device’s fundamental operations. Gate Set Tomography relaxes the strict SPAM assumptions of QPT since it does not assume that state prep or measurement is known or ideal. Instead, it treats them as unknown processes to be learned. However, GST still relies on the assumption that errors are Markovian and time-invariant during the experiment; otherwise, it is unable to learn about the SPAM errors through its modelling. By repeating gates in long sequences, one can amplify small errors linearly without encountering new error modes, which works especially well for coherent errors [19].

Scalability is limited, just as was the case in QPT. In GST, the number of parameters explodes with qubit count, making GST on anything over 2-3 qubits practically infeasible at the moment. In fact, there are even extra classical algorithm burdens on GST as it requires solving a large non-linear optimization problem to fit the gate set to observed frequencies [20].

GST amplifies systematic coherent errors through long sequences, making even small coherent rotations visible as a measurable deviation, which appears linearly in the measurement data [18]. GST is equally sensitive to stochastic errors. It can also detect cross-talk or correlated errors between gates if the experiments are designed appropriately, since any consistent effect needed to explain the data will be captured in the fitted gate set. Additionally, GST includes internal checks for model adequacy, so if there are non-Markovian or non-stationary effects that the model cannot capture, the GST log-likelihood will indicate a poor fit [21].

GST provides detailed and actionable error models for individual quantum gates, allowing researchers to debug and pinpoint specific error sources like misalignment or calibration drifts, and improve gates by adjusting controls or hardware. Its accurate estimates are useful for validating fault-tolerance thresholds for quantum error correction. Although GST is computationally expensive and typically used on small subsets of a processor, its detailed

diagnostics offer a clear roadmap for improving gate fidelity and can be trusted even in the presence of SPAM errors.

Direct Fidelity Estimation

DFE is a targeted protocol that estimates the fidelity between an experimentally realized quantum state (or process) and its desired ideal state, using far fewer measurements than full tomography. The key idea is to leverage prior knowledge of the ideal state or gate, and instead of reconstructing the entire density matrix or process like QFT, DFE selectively measures certain observables (often Pauli operators) whose expected values in the ideal state are known [22]. DFE yields a single metric called fidelity, which quantifies the accuracy of the state or operation in question. In other words, the result is a single numerical value reflecting how closely the implemented operation or prepared state approximates the intended one.

DFE assumes that the target state or operation is fully specified in advance and that the quantum system permits accurate measurements in the required bases. It critically depends on the ability to perform projective measurements, usually in the Pauli basis, and assumes that these, along with state preparation procedures, are either error-free or contribute minimally to the overall fidelity loss. There is no inherent mechanism in DFE to decouple SPAM errors from gate or process errors, which means that all errors contribute indistinguishably to a lower fidelity estimate [23]. Additionally, DFE assumes that the system noise is stationary and Markovian, so the fidelity estimate remains consistent across repeated trials. Without this, the measured fidelities may fluctuate, and the reliability of DFE would deteriorate. In short, the assumptions of DFE are nearly identical to those of QPT.

The efficiency of DFE is one of its primary advantages. Unlike full quantum state estimation or process tomography, which scales exponentially in the number of qubits, DFE requires only a number of measurements that is polynomial or even independent of system size for fixed fidelity estimation accuracy [24]. This makes DFE practical for use in larger-scale quantum devices, especially where full tomography is infeasible. However, this efficiency is contingent on having a sparsely decomposable target state or process in the Pauli basis and access to efficient measurement strategies, which may not always be attainable for quantum systems.

DFE does not provide a resolution into the type or source of the error, as it captures only the cumulative impact in a single number. Importantly, because SPAM errors are indistinguishable from gate errors in DFE's framework, these can distort results unless separately

accounted for. Coherent errors may also be underreported, particularly if they result in states that retain high overlap with the target state.

The primary actionability of DFE lies in its role as a fast, scalable fidelity certifier. DFE is often used as a cost function in gate calibration routines: by optimizing control parameters to maximize fidelity with a known target, devices can be tuned for better performance [23]. However, because DFE does not reveal the origin of fidelity loss, its role is typically limited to performance validation and comparative benchmarking.

While general benchmarking techniques provide a comprehensive view of gate performance, they also reveal limitations related to scalability and sensitivity that motivate the development of new methods. Additionally, these approaches have been assumed to be confined until now. To address these drawbacks, we will next shift our view to the two emerging benchmarking methods: Leakage Benchmarking and Deterministic Benchmarking.

Leakage Benchmarking

Leakage refers to the unwanted population of states outside the intended computational subspace, typically beyond the $|0\rangle$ and $|1\rangle$ levels that define a qubit. Leakage errors are particularly detrimental because they violate the basic assumption underlying most error correction and benchmarking schemes: that errors remain confined to a two-level system. While traditional benchmarking methods such as randomized benchmarking (RB) or interleaved RB quantify average gate infidelity within the computational subspace, they are generally insensitive to population transfer into leakage states. Consequently, dedicated leakage benchmarking techniques have been developed to characterize this distinct class of errors.

Leakage benchmarking methods extend the randomized benchmarking framework to measure both leakage and seepage rates. The leakage rate quantifies the probability that the population leaves the computational subspace during gate application, while the seepage rate describes the probability that leaked population returns. The most common implementation applies a sequence of randomly sampled Clifford gates, identical to standard RB, but augments the analysis by measuring the total survival probability in the computational subspace. By fitting the decay of this survival probability as a function of sequence length, one can extract the leakage and seepage parameters independently of depolarizing noise assumptions. This approach was introduced and formalized by Wood and Gambetta [25], providing an operational way to separate leakage from other stochastic errors.

In practice, leakage benchmarking has been applied across multiple hardware platforms, including superconducting qubits and trapped ions, where higher excited states or motional modes can participate unintentionally in gate dynamics. In superconducting systems, for example, strong microwave drives can populate the $|2\rangle$ level of a transmon, leading to leakage that accumulates over many gates and limits the effectiveness of error correction. Leakage benchmarking provides a means of quantifying such effects without requiring full state tomography, thus enabling scalable characterization of higher-level population dynamics.

More recent advancements have focused on interleaved leakage benchmarking, which isolates leakage contributions from specific gate implementations, and subspace tomography-based approaches that complement leakage RB with spectroscopic identification of leaked populations. Together, these methods contribute to a more complete understanding of quantum gate performance by capturing errors that standard RB or process tomography may obscure. As quantum processors scale, leakage benchmarking will remain essential for ensuring that qubit control remains confined to the intended computational manifold, preserving the validity of quantum error correction assumptions and overall gate fidelity metrics.

Deterministic Benchmarking

While randomized benchmarking (RB) remains the predominant framework for estimating average gate fidelities in quantum processors, it inherently measures statistical averages over randomly sampled sequences. This randomness, while powerful for isolating depolarizing noise, obscures fine-grained and deterministic error structures such as coherent over-rotations, calibration drifts, and crosstalk. To overcome these limitations, deterministic benchmarking methods have been developed to evaluate gate performance using fixed, non-randomized sequences that reveal systematic error mechanisms with higher resolution.

Deterministic benchmarking employs gate sequences whose outcomes are analytically predictable in the absence of noise. Instead of averaging over random Clifford operations, these protocols apply deterministic patterns. These are often based on mutually unbiased bases or specific group decompositions and are intended to probe gate errors directly. This approach enables the characterization of coherent and correlated errors that randomized schemes tend to average out. For example, deterministic benchmarking can identify small phase or amplitude errors accumulated across repeated gates. Applications for this are in unitary error detection, where accumulation would otherwise appear as stochastic noise under RB.

One representative deterministic method is Cycle Benchmarking, which applies a fixed cycle of Clifford gates repeatedly to measure how errors compound over multiple applications. By analyzing the deterministic decay of expectation values under specific input states, cycle benchmarking can isolate gate-set dependent errors and yield average process fidelities without requiring randomization. Another deterministic variant, Symmetrized Characterization of Unitaries (SCU), constructs fixed gate sequences that amplify coherent miscalibrations in specific channels, thus providing a diagnostic tool for identifying and correcting systematic gate errors.

Deterministic benchmarking protocols also include gate set tomography (GST) and Hamiltonian learning-based methods, which, although more resource-intensive, provide deterministic reconstructions of error channels from well-defined experimental sequences. Unlike RB, these methods can disentangle different types of coherent errors, such as control miscalibrations or non-Markovian noise, at the cost of greater experimental and computational overhead.

The primary advantage of deterministic benchmarking lies in its interpretability and diagnostic precision. Because the gate sequences and expected outcomes are explicitly known, deterministic methods offer clearer mappings between observed deviations and physical error sources. However, the absence of statistical averaging also makes these methods more susceptible to state preparation and measurement (SPAM) errors, which randomized approaches naturally suppress. Consequently, deterministic benchmarking is most effective when combined with RB-style techniques.

Conclusion

Quantum gate benchmarking is essential for improving the performance of quantum computing-based technologies. The simple nature of randomized benchmarking, while providing scalable average-fidelity estimates, leads to its popularity, but it can miss coherent and leakage errors, but novel sequences are actively being developed to help quantify what randomized benchmarking can fail to characterize.

More detailed methods, such as QPT, GST, and DFE, offer deeper insights but are inherently limited by scalability and SPAM errors. Leakage benchmarking and deterministic benchmarking address issues with prevailing methods surrounding detecting population outside of allowable subspace and systematic errors, respectively. From a comprehensive standpoint, these methods provide pathways toward scalable quantum technologies.

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